

Trainings ▾

Install

Use clean engine oil to lubricate the oil pressure regulator bores on the flywheel housing.

Install the pressure regulator plunger (3) into the bore with the flat face of the plunger first.

Check that the plunger has freedom of movement the entire length of the bore.

Insert the regulator valve springs (4) into the bore.

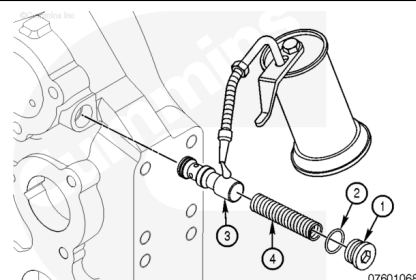
Assemble a new o-ring (2) on the plug (1).

Lubricate the o-ring (2) and compress the pressure regulator spring sufficiently to engage the plug to the threads in the bore. Tighten the plug.

Torque Value:

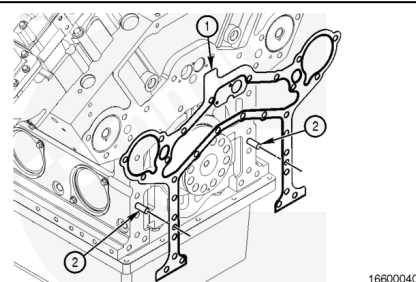
Oil Pressure Regulator Plug

1. 60 n•m [44 ft-lb]



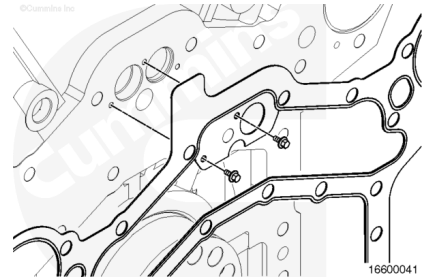
⚠ CAUTION ⚠

The tab at the top of the flywheel housing gasket is to provide indication of proper orientation of the gasket. Failure to do so will cause excessive lubrication oil temperatures which will lead to engine damage.



Install the flywheel housing gasket over the dowels (2) on the engine block. Make sure that the tab (1) on the gasket is located on the same side as the lubricating oil filter head assembly. The tab shows which of the two oil cooler rifles will be covered to make sure oil is passed through the oil coolers. Example: If the oil filter head is on the right bank of the engine, the right bank oil cooler rifle needs to be covered. Therefore, the tab (1) on the flywheel housing gasket must be on the right bank side.

Secure the flywheel gasket in place with the two capscrews located below the tab.



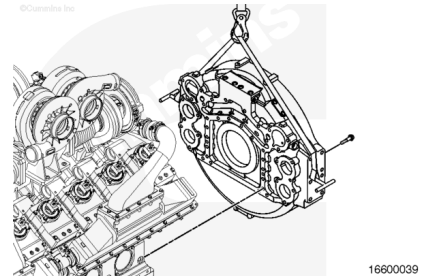
Torque Value:

Flywheel Gasket Capscrews

1.9 n·m [80 in-lb]

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Make sure the flywheel housing dowel pins have been installed into the cylinder block.

Use guide bolts to help during alignment.

Install the flywheel housing and the capscrews.

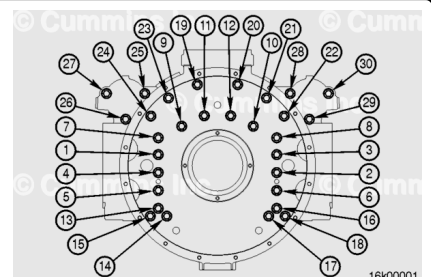
Tighten the flywheel housing mounting capscrews in sequence.

Torque Value:

1. 95 n·m [70 ft-lb]

2. 195 n·m [144 ft-lb]

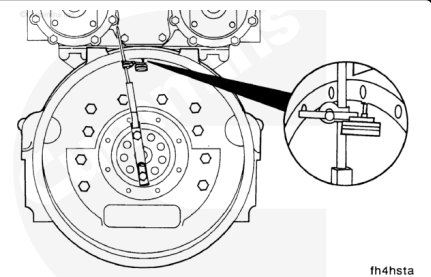
Note : There are two types of flywheel housings used on QSK45 and QSK60 engine applications; small (0) and large (00). A small (0) flywheel housing is shown in the illustration. The torque sequence and torque values apply to both flywheel housings.



The bore and the face of the housing **must** align with the crankshaft.

The indicator arm **must** be rigid for an accurate reading. It **must not** sag.

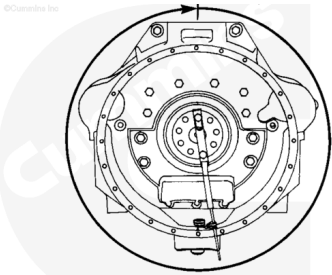
Attach an indicator to the crankshaft as shown.



Position the indicator at the 12-o'clock position. Adjust the dial until the needle points to zero. Rotate the crankshaft one complete revolution, 360 degrees.

Record the total indicator runout.

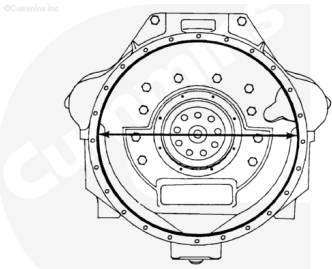
The indicator **must** return to zero when returning to the start point.



fh4hstb

The maximum allowable total indicator runout depends on the diameter of the bore.

Flywheel Housing Bore Diameter			
	mm		in
Number 0	648	NOM	25.5
Number 00	787	NOM	31.0



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Flywheel Housing - Radial Runout			
	mm		in
Number 0		0.25	MAX
Number 00		0.30	MAX

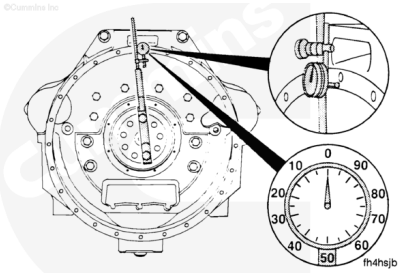
If the alignment is **not** within specifications and the bore is round, the housing can be shifted.

If the alignment is **not** within specifications and the bore is **not** round, the housing **must** be replaced.

When measuring the face runout, the crankshaft **must** be pushed or pulled in the same direction each time a point is measured.

Attach an indicator as shown. Position the indicator at the 12-o'clock position. Adjust the dial until the needle points to zero.

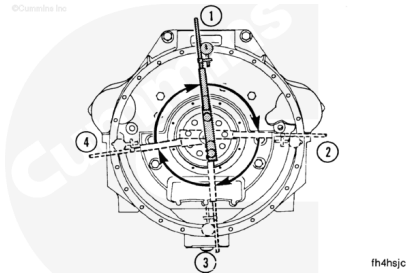
Flywheel Housing - Face Runout			
	mm		in
Number 0		0.25	MAX
Number 00		0.30	MAX



fh4hsjb

Record the indicator reading at three different points; for example, 3 o'clock, 6 o'clock, and 9 o'clock.

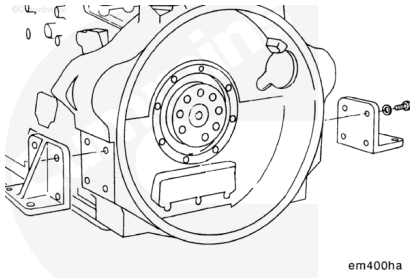
Turn backward to the original position. Make sure the needle still points to zero. Determine the total indicator runout.



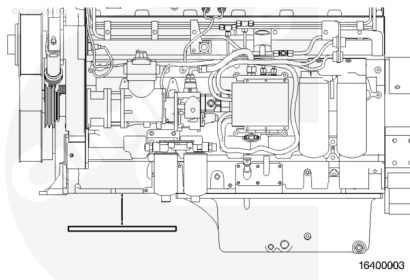
Example of Total Indicator Run Out	
Position	Indicator Reading
3 o'clock	0.00 mm [0.00 in]
6 o'clock	-0.08 mm [-0.003 in]
9 o'clock	0.05 mm [0.002 in]
Total Indicator Run Out = 0.13 mm [0.005 in]	

Install the rear engine mounts to the flywheel housing.

Torque Value: 550 n•m [406 ft-lb]



Install the adapter cover plate or oil sump, whichever is in the front position.



Lift the engine. Remove the jack stands and supports, and lower the engine.

